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Agent vs LLM

LLM (Large Language Model)

What it is: A language model trained on massive amounts of text, such as GPT-4, BERT, LLaMA, etc.

Main function: To generate text, complete instructions, answer questions, translate, summarize, etc.

Nature: Reactive. It responds to input (prompts) without memory or self-initiative unless integrated into a system that provides that.

Does not have: Goals, external tools, or persistent memory on its own.

Agent (AI Agent)

What it is: An autonomous system that uses an LLM (or other models) as its reasoning engine, but:

- Has defined goals.
- Can plan intermediate steps.
- Has access to external tools (such as web browsing, code execution, database queries).
- May have long-term memory.

Main function: To solve complex, dynamic tasks by acting in open-ended environments.

Examples: AutoGPT, BabyAGI, LangGraph agents, or agents in platforms like ChatGPT when tools are enabled.

Relationship between the two

An agent can be built on top of an LLM, but not all LLMs are agents. The LLM is the linguistic and reasoning core, while the agent is an orchestrator of actions.

Analogy

LLM: Like an expert who answers any question, but only if you ask.

Agent: Like an autonomous assistant who can investigate, ask questions, make decisions, and act on its own to achieve a goal.
